PHYS 8602

Problem #1 Due 1/27/25

Consider a 1-dim, simple harmonic oscillator with Hamiltonian

$$\mathcal{H} = \frac{p^2}{2m} + \frac{kx^2}{2},$$

where p is the momentum, x is the position, m is the mass, and k is the spring constant. The total energy E is constant, in theory. Set $k = 1$, $m = 1$ with initial values $v_0 = 0$, $t_0 = 0$, and $x_0 = 1$ (in appropriate units) and integrate the equation(s) of motion for this system using time integration intervals $\Delta = 0.001, 0.01$, and 0.1 from time $t = 0$ out to $t_{max} = 10$. You are to run your orogram on *TeachingCluster* and submit a brief report that includes results and analysis with the following information:

1.) Plot the error in E and x as a function of Δ for suitable times.

2.) Plot p vs x for values of t between $t = 0$ and $t = t_{max}$.

3.) Comment on your results in parts 1.) and 2.).

Please make your report concise but describe how you analyzed your results and how you reached your conclusions. Include an annotated copy of your program and a copy of a summary.log.

Your report should be submitted as a single pdf file to dlandau@uga.edu. *The filename should be "your_last name"_PHYS8602_prob1.pdf. Please remember to put your name on your report!*